



Hitachi Portal Integration (Android)

For Internal Use Only

Hitachi Portal Integration (Android)

Confidentiality

This document contains proprietary information that is confidential to Hitachi Asia. Disclosure of this document in full or in part, may result in material damage to Hitachi Asia. Written permission must be obtained from Hitachi Asia prior to the disclosure of this document to a third party.

Version: 0.6 **File Name:** Hitachi Portal Integration (Android)_v0.6 **Confidentiality Category:** For Internal Use Only

Document Change Control

Version	Date	Author(s)	Description
0.1	15 July 2024	Chris Lee	Initial Version
0.2	30 July 2024	Chris Lee	Update addEventListener
0.3	6 Aug 2024	Chris Lee	Update Notify Listener
0.4	17 Oct 2024	Chris Lee	Update Permission Required and Additional Configuration for Portal Open
0.5	22 Oct 2024	Chris Lee	onRequestPermissionsResult and onActivityResult
0.6	5 Mar 2026	Chris Lee	Add Geolocation support

Table of Contents

[Introduction](#)

[1.1 Overview](#)

[1.2 What's New in v0.6](#)

[Specification](#)

[Installation](#)

[3.1 Add the AAR Module to Your Project](#)

[3.2 Update Build Settings](#)

[3.3 Permission Required](#)

[Example Usage](#)

[4.1 Basic Portal](#)

[4.2 Geolocation](#)

[API](#)

[5.1 Portal](#)

[5.2 open](#)

[5.3 close](#)

[5.4 addEventListener](#)

[5.5 notifyListener](#)

1. Introduction

1.1 Overview

This document provides a comprehensive overview of the HASPortal SDK, highlighting its primary features and benefits for Android development. The SDK includes functionalities such as Wrapping, Token Interceptor, and Publish-Subscribe to enhance the overall performance, security, and scalability of the application.

Objectives:

- **Encapsulation and Security:** The SDK ensures that the portal functionality is encapsulated within a secure environment, providing better control over lifecycle events and consistent integration across different parts of the application.
- **Enhanced Security and Streamlined Authentication:** The Token Interceptor automates the management of authentication tokens, simplifying user session handling and ensuring secure communication between the app and backend services.
- **Asynchronous Communication and Decoupling:** The Pub-Sub mechanism facilitates non-blocking, event-driven communication, promoting a modular and scalable architecture by decoupling event producers and consumers.

Key Benefits:

- **Encapsulation:** Keeps portal interactions secure and consistent, with enhanced control over initialization, usage, and termination.
- **Security and Efficiency:** Token management, reducing the risk of errors and ensuring secure network communications.
- **Modularity and Flexibility:** Supports dynamic addition and removal of event listeners, improving application responsiveness and maintainability through asynchronous event handling.

1.2 What's New in v0.6

1. Geolocation Support

2. Web content can now use `navigator.geolocation` to access device location
 3. The SDK automatically handles the WebView geolocation permission delegate
(`onGeolocationPermissionsShowPrompt`)
 4. Host app needs to add `ACCESS_FINE_LOCATION` (required) and `ACCESS_COARSE_LOCATION` (optional) permissions to `AndroidManifest.xml`
 5. Runtime permission prompt is handled automatically by the SDK
 6. No custom event handling code required — the standard browser Geolocation API works directly
-

2. Specification

- **Android minimum SDK:** 24
 - **SDK Size:** 27kb
-

3. Installation

3.1 Add the AAR Module to Your Project

- Open your project in Android Studio.
- Place the AAR file in the `app/libs` directory of your project.
- Create the `libs` directory inside `app` if it does not exist.

Project structure:

```
android [My Application]
├── .gradle
├── .idea
├── app
│   ├── build
│   ├── libs
│   │   └── HASPortal-release.aar
│   ├── src
│   ├── .gitignore
│   ├── build.gradle.kts
│   └── proguard-rules.pro
└── build
```

3.2 Update Build Settings

- Open the **build.gradle** / **build.gradle.kts** file of your app.
- Add the following lines to the dependencies section to include the AAR module:

For build.gradle:

```
dependencies {
    implementation("androidx.appcompat:appcompat:1.4.1")
    implementation("androidx.browser:browser:1.4.0")
    implementation("com.google.android.material:material:1.5.0")

    implementation files("libs/HASPortal-release.aar")
}
```

For build.gradle.kts:

```
dependencies {
    implementation("androidx.appcompat:appcompat:1.4.1")
    implementation("androidx.browser:browser:1.4.0")
    implementation("com.google.android.material:material:1.5.0")
}
```

```
implementation(files("libs/HASPortal-release.aar"))
}
```

3.3 Permission Required

- Add the following permissions for SDK usage
- Skip if the permissions are added

AndroidManifest Requirements:

```
<uses-feature android:name="android.hardware.camera" android:required="false" />
<uses-permission android:name="android.permission.CAMERA"/>
<uses-permission android:name="android.permission.INTERNET"/>

<!-- Required for geolocation -->
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<!-- Optional: allows approximate location as a fallback -->
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
```

Note: `ACCESS_FINE_LOCATION` is **required** for the SDK's geolocation feature to work. Without it, the runtime permission request will fail silently and the WebView will not receive location data.

`ACCESS_COARSE_LOCATION` is optional and provides approximate location as a fallback.

Add `onRequestPermissionsResult` and `onActivityResult` for permission result handling for `ComponentActivity`:

```
class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
    }

    override fun onRequestPermissionsResult(
        requestCode: Int,
        permissions: Array<out String>,
        grantResults: IntArray
    ) {
        portal.onRequestPermissionsResult(requestCode, permissions, grantResults)
    }

    override fun onActivityResult(requestCode: Int, resultCode: Int, data: Intent?) {
        super.onActivityResult(requestCode, resultCode, data)

        // Forward to Portal if it exists
        if (::portal.isInitialized) {
            portal.handleActivityResult(requestCode, resultCode, data)
        }
    }
}
```

4. Example Usage

4.1 Basic Portal

Below is an example of how to use the integrated SDK in your Android project:

```
package com.has.myapplication

import android.content.Context
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.fillMaxSize
import androidx.compose.material3.Button
import androidx.compose.material3.MaterialTheme
import androidx.compose.material3.Surface
import androidx.compose.material3.Text
import androidx.compose.runtime.Composable
import androidx.compose.ui.platform.LocalContext
import com.has.portal.Portal

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            Surface(color = MaterialTheme.colorScheme.background) {
                PortalButton()
            }
        }
    }
}

@Composable
fun PortalButton() {
    val context = LocalContext.current
    val activity = context as? ComponentActivity

    Button(
        onClick = { activity?.let { openPortal(context, it, "https://example.com") } },
        modifier = Modifier.padding(8.dp)
    ) {
        Text("Loyalty Program")
    }
}

fun openPortal(context: Context, activity: ComponentActivity, url: String) {
    val configuration: MutableMap<String, Any> = mutableMapOf(
        "accessToken" to "access token",
    )

    val portal = Portal(context, activity, configuration)

    portal.addEventListener("eventsCallbacks") { data ->
        println("Listener event data: $data")

        val name = data["name"] as? String
        val type = data["type"] as? String

        when {
            name == "redeem" && type == "navigation" -> {
                println("Handling redeem navigation event")
                portal.close()
                // Handle the redeem navigation event here
            }
        }
    }
}
```

```

        name == "close" -> {
            println("Handling close event")
            // Handle the close event here
        }
        name == "token-expired" -> {
            println("Handling Invalid Token event")
            // Handle the access token expired event here
            val newConfiguration: MutableMap<String, Any> = mutableMapOf(
                "accessToken" to "new token",
            )
            portal.notifyListener("token-updated", newConfiguration)
        }
    }
    null
}
portal.open(url)
}

```

4.2 Geolocation

The SDK supports `navigator.geolocation` in web content loaded within the portal. The SDK handles the WebView geolocation delegate automatically — no native event code is required from the host app.

Host App Setup

Add location permissions to your `AndroidManifest.xml` (see [Section 3.3](#)).

The `onRequestPermissionsResult` override (already required for camera) automatically forwards location permission results to the SDK. No additional native code is needed.

Web-side Code (JavaScript)

Use the standard browser Geolocation API — no custom events required:

```

// Get current position
navigator.geolocation.getCurrentPosition(
    function(position) {
        console.log('Latitude:', position.coords.latitude);
        console.log('Longitude:', position.coords.longitude);
        console.log('Accuracy:', position.coords.accuracy, 'meters');
    },
    function(error) {
        console.error('Geolocation error:', error.message);
    },
    {
        enableHighAccuracy: true,
        timeout: 10000,
        maximumAge: 0
    }
);

// Watch position (continuous updates)
var watchId = navigator.geolocation.watchPosition(
    function(position) {
        console.log('Updated position:', position.coords.latitude, position.coords.longitude);
    },
    function(error) {
        console.error('Watch error:', error.message);
    }
);

```



```
// Stop watching
navigator.geolocation.clearWatch(watchId);
```

Permission Flow: 1. Web content calls `navigator.geolocation.getCurrentPosition()` or `watchPosition()` 2. The SDK's `onGeolocationPermissionsShowPrompt` is triggered 3. If permission already granted: location data flows directly to the web callback 4. If not granted: the SDK requests `ACCESS_FINE_LOCATION` at runtime via a headless fragment 5. If denied: the SDK shows a Settings redirect dialog 6. Permission result is forwarded back to the WebView geolocation callback

Important Notes: - `ACCESS_FINE_LOCATION` **must** be declared in `AndroidManifest.xml`, otherwise the runtime permission request will fail silently - Unlike contacts and other custom permissions, geolocation does **not** use the custom event system (`cap-events`). It works through the standard `navigator.geolocation` browser API - The SDK only requests `ACCESS_FINE_LOCATION` at runtime. If your app only declares `ACCESS_COARSE_LOCATION`, the SDK's permission check will not match and geolocation will be denied

5. API

5.1 Portal

```
public constructor(context: Context, activity: Activity, configuration: MutableMap<String, Any>)
```

Parameters

Param	Type	Additional Information
context	Context	The context of the application.
activity	ComponentActivity	The activity from which the portal is launched.
configuration	MutableMap<String, Any>	A mutable map containing configuration options such as <code>accessToken</code> , <code>title</code> and <code>showHeader</code> .

Sample

```
val configuration: MutableMap<String, Any> = mutableMapOf(
    "accessToken" to "access token",
    "title" to "Portal", // optional
    "showHeader" to "true", // optional
)

val portal = Portal(context, activity, configuration)
```

5.2 open

```
public void open(String urlString)
```

Parameters

Param	Type	Additional Information
urlString	String	

Sample

```
var urlString = "https://example.com"
portal.open(url)
```

5.3 close

```
public void close()
```

Sample

```
portal.close()
```

5.4 addEventListener

```
public void addEventListener(String eventName, Function<Map<String, Object>, Map<String, Object>> listener)
```

Parameters

Param	Type	Additional Information
eventName	String	
listener	Consumer<Map<String, Object>>	

Sample

```
portal.addEventListener("eventsCallbacks") { data ->
    println("Listener event data: $data")

    val name = data["name"] as? String
    val type = data["type"] as? String

    when {
        name == "redeem" && type == "navigation" -> {
            println("Handling redeem navigation event")
            portal.close()
            // Handle the redeem navigation event here
        }
        name == "close" -> {
            println("Handling close event")
            // Handle the close event here
        }
        name == "token-expired" -> {
            println("Handling Invalid Token event")
            // Handle the access token expired event here
            val newConfiguration: MutableMap<String, Any> = mutableMapOf(
                "accessToken" to "new token",
```

```
        )
        portal.notifyListener("token-updated", newConfiguration)
    }
}
null
}
```

5.5 notifyListener

```
private void notifyListener(String eventName, Map<String, Object> data)
```

Parameters

Param	Type	Additional Information
eventName	String	
data	Map<String, Object>	

Sample

```
val newConfiguration: MutableMap<String, Any> = mutableMapOf(
    "accessToken" to "new token",
)
portal.notifyListener("token-updated", newConfiguration)
```

Appendix: Migration from v0.5 to v0.6

If you're upgrading from version 0.5, here are the key changes:

New Features

1. **Geolocation Support** - Web content can use `navigator.geolocation` to access device location

AndroidManifest Requirements

Add the following permissions if your web content uses geolocation:

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
```

Breaking Changes

None - v0.6 is fully backward compatible with v0.5

Recommended Updates

1. Add `ACCESS_FINE_LOCATION` to `AndroidManifest.xml` if your web content uses geolocation

2. No native code changes required — the SDK handles geolocation permission automatically via the existing `onRequestPermissionsResult` forwarding
-

End of Document